

Hands-on Activity 6.1 | Ansible Roles

Reference:

```
---
- hosts: all
  become: true
  tasks:

    - name: update repository index
      apt:
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

    - name: install apache2 package
      apt:
        name: apache2
        state: latest
      when: ansible_distribution == "Ubuntu"

    - name: add PHP support for apache
      apt:
        name: libapache2-mod-php
        state: latest
      when: ansible_distribution == "Ubuntu"

    - name: update repository index
      dnf:
        update_cache: yes
      when: ansible_distribution == "CentOS"

    - name: install apache2 package
      dnf:
        name: httpd
        state: latest
      when: ansible_distribution == "CentOS"

    - name: add PHP support for apache
      dnf:
        name: php
        state: latest
      when: ansible_distribution == "CentOS"
```

Creating Roles

```
jalene@workstation:~/CPE212_jalene$ ansible-galaxy init ubuntu_apache --offline
- Role ubuntu_apache was created successfully
jalene@workstation:~/CPE212_jalene$ ansible-galaxy init centos_apache --offline
- Role centos_apache was created successfully
jalene@workstation:~/CPE212_jalene$ ls
ansible.cfg      gateway.yml      playbook.yml      ubuntu_apache
apache           install_apache.yml  README.md
centos_apache    inventory.ini      Santiago_PrelimExam

jalene@workstation:~/CPE212_jalene$ ansible-galaxy init mariadb --offline
- Role mariadb was created successfully
jalene@workstation:~/CPE212_jalene$
```

ubuntu_apache main.yml

```
EXPLORER  ...  ! main.yml ubuntu_apache/... U X  ! main.yml mariadb/... U  ! main.yml centos_apache/... U  ! playbook2.yml U

CPE212_JALENE
├── centos_apache
├── mariadb
├── ubuntu_apache
│   ├── defaults
│   ├── files
│   ├── handlers
│   ├── meta
│   └── tasks
│       ├── ! main.yml U
│       ├── templates
│       ├── tests
│       ├── vars
│       └── README.md U
├── ansible.cfg
├── ! install_apach... IA, U
├── inventory.ini M
├── ! playbook.yml
├── ! playbook2.yml U
└── Santiago_Prelim... U

ubuntu_apache > tasks > ! main.yml
1  - name: Ubuntu Installation
2    apt:
3      name: apache2
4      state: latest
5    become: yes
6    when: ansible_distribution == "Ubuntu"
7
8  - name: Update Repository Index
9    apt:
10     update_cache: yes
11   become: yes
12   when: ansible_distribution == "Ubuntu"
13
14 - name: Add PHP support for apache
15   apt:
16     name: libapache2-mod-php
17     state: latest
18   become: yes
19   when: ansible_distribution == "Ubuntu"
```

centos_apache main.yml

```
EXPLORER  ...  ! main.yml ubuntu_apache/... U  ! main.yml mariadb/... U  ! main.yml centos_apache/... U X  ! playbook2.yml U

CPE212_JALENE
├── centos_apache
│   ├── defaults
│   ├── files
│   ├── handlers
│   ├── meta
│   └── tasks
│       ├── ! main.yml U
│       ├── templates
│       ├── tests
│       ├── vars
│       └── README.md U
├── mariadb
│   ├── defaults
│   ├── files
│   ├── handlers
│   ├── meta
│   └── tasks
│       ├── ! main.yml U
│       └── templates
├── ansible.cfg
├── ! install_apach... IA, U
├── inventory.ini M
├── ! playbook.yml
├── ! playbook2.yml U
└── Santiago_Prelim... U

centos_apache > tasks > ! main.yml
1  - name: Ubuntu Installation
2    apt:
3      name: apache2
4      state: latest
5    become: yes
6    when: ansible_distribution == "CentOS"
7
8  - name: Update Repository Index
9    apt:
10     update_cache: yes
11   become: yes
12   when: ansible_distribution == "CentOS"
13
14 - name: Add PHP support for apache
15   apt:
16     name: libapache2-mod-php
17     state: latest
18   become: yes
19   when: ansible_distribution == "CentOS"
```

mariadb main.yml

```
EXPLORER  ...  ! main.yml ubuntu_apache/... U  ! main.yml mariadb/... U X  ! main.yml centos_apache/... U  ! playbook2.yml U

CPE212_JALENE
├── centos_apache
│   ├── defaults
│   ├── files
│   ├── handlers
│   ├── meta
│   └── tasks
├── mariadb
│   ├── defaults
│   ├── files
│   ├── handlers
│   ├── meta
│   └── tasks
├── ansible.cfg
├── ! install_apach... IA, U
├── inventory.ini M
├── ! playbook.yml
├── ! playbook2.yml U
└── Santiago_Prelim... U

mariadb > tasks > ! main.yml
1  - name: MariaDB Installation
2    apt:
3      name: mariadb-server
4      state: present
5      update_cache: yes
6    become: yes
```

playbook2.yml

```
! main.yml ubuntu_apache/... U    ! main.yml mariadb/... U    ! main.yml centos_apache/... U    ! playbook2.yml U X

! playbook2.yml
1  - name: Apache 2
2    hosts: all
3    become: true
4
5    roles:
6      - ubuntu_apache
7      - mariadb
```

I did not include the role for CentOS anymore as my CentOS is currently not working so running it will just cause an error.

OUTPUT

Running both apache and mariadb Roles

```
File Edit Selection View Go ... CPE212_jalene
! main.yml ubuntu_apache/... U    ! main.yml mariadb/... U    ! main.yml centos_apache/... U    ! playbook2.yml U X

! playbook2.yml
6    - ubuntu_apache
7    - mariadb

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● jalene@workstation:~/CPE212_jalene$ ansible-playbook playbook2.yml -K
BECOME password:

PLAY [Apache 2] *****

TASK [Gathering Facts] *****
ok: [server2]
ok: [server1]

TASK [ubuntu apache : Ubuntu Installation] *****
ok: [server1]
ok: [server2]

TASK [ubuntu apache : Update Repository Index] *****
changed: [server1]
changed: [server2]

TASK [ubuntu apache : Add PHP support for apache] *****
ok: [server2]
ok: [server1]

TASK [mariadb : MariaDB Installation] *****
ok: [server1]
ok: [server2]

PLAY RECAP *****
server1      : ok=5    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2      : ok=5    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

○ jalene@workstation:~/CPE212_jalene$
```

Running only mariadb Role

```
! main.yml ubuntu_apache/... U  ! main.yml mariadb/... U  ! main.yml centos_apache/... U  ! playbook2.yml U X
```

```
! playbook2.yml
1  - name: Apache 2
2    hosts: all
3    become: true
4
5    roles:
6      - mariadb
```

```
PROBLEMS  DEBUG CONSOLE  OUTPUT  TERMINAL  PORTS
```

```
● jalene@workstation:~/CPE212_jalene$ ansible-playbook playbook2.yml -K
BECOME password:

PLAY [Apache 2] *****

TASK [Gathering Facts] *****
ok: [server2]
ok: [server1]

TASK [mariadb : MariaDB Installation] *****
ok: [server2]
ok: [server1]

PLAY RECAP *****
server1                : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

○ jalene@workstation:~/CPE212_jalene$
```